

KREMSMUNSTER, AustriaWMO: **110120**Lat: **48.0550N**Lon: **14.1311E**Elev: **390**StdP: **96.73**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-8.7	-13.0	1.3	-10.4	-10.4	1.6	-7.5	12.1	4.6	10.9	4.5	2.7	230	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	30.5	20.7	28.5	20.0	26.9	19.2	21.8	28.6	20.8	27.4	19.9	26.1	2.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	15.0	25.4	18.5	14.0	24.1	17.6	13.2	23.1	65.6	28.6	61.9	27.4	58.6	26.3	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.5	7.5	-13.1	33.8	3.3	1.7	-15.5	35.0	-17.4	35.9	-19.3	36.9	-21.6	38.1		
(5)			-13.6	23.5	3.1	0.9	-15.8	24.1	-17.6	24.6	-19.4	25.1	-21.6	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-0.8	1.2	5.4	10.6	14.7	18.2	19.9	19.6	15.0	10.0	4.6	0.7		
	DBStd	8.28	4.63	4.57	4.22	4.32	3.69	3.76	3.32	3.44	3.30	3.71	3.83	4.15		
	HDD10.0	1290	335	248	149	45	4	1	0	0	2	48	168	290		
	HDD18.3	3254	593	480	400	233	122	48	23	26	109	259	413	548		
	CDD10.0	1278	1	1	8	63	152	246	307	298	152	47	5	0		
	CDD18.3	201	0	0	0	1	11	44	72	66	8	0	0	0		
	CDH23.3	1504	0	0	0	15	88	333	538	478	51	1	0	0		
	CDH26.7	413	0	0	0	1	11	84	162	148	6	0	0	0		
(14) Wind	WSAvg	3.4	3.7	3.7	3.9	3.5	3.4	3.1	3.3	3.0	3.1	3.1	3.2	3.4		
(15) Precipitation	PrecAvg	955	58	58	63	76	100	114	128	120	73	59	72	60		
	PrecMax	1343	112	151	136	184	215	176	233	358	114	182	153	130		
	PrecMin	717	4	6	16	25	37	50	42	30	27	3	26	10		
	PrecStd	167	32	36	29	43	46	39	49	67	23	43	32	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	13.9	19.0	25.3	28.2	31.4	33.1	33.4	27.3	22.0	16.8	12.0		
		MCWB	8.7	8.6	11.9	15.6	18.6	21.5	20.8	20.8	19.5	15.8	11.6	8.6		
	2%	DB	9.9	11.1	16.1	22.4	25.8	29.2	30.4	30.2	24.7	18.9	13.7	9.7		
		MCWB	7.2	7.5	10.4	14.2	17.6	20.6	20.5	20.6	18.2	14.6	10.6	7.4		
	5%	DB	7.4	9.1	13.8	20.1	23.7	27.2	28.3	28.0	22.5	16.9	11.5	7.7		
		MCWB	5.4	6.2	9.2	13.3	16.4	19.4	19.8	20.1	17.3	13.6	9.6	5.8		
	10%	DB	5.4	7.2	11.6	17.9	21.5	25.1	26.4	26.0	20.5	15.2	9.7	6.0		
		MCWB	3.9	4.9	8.0	12.1	15.3	18.4	19.1	19.4	16.3	12.8	8.1	4.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	9.4	13.0	16.5	19.6	23.0	23.1	23.5	20.5	16.7	12.5	9.3		
		MCDB	11.9	13.1	16.4	23.2	26.2	29.6	29.3	28.6	25.8	20.5	15.2	11.4		
	2%	WB	7.3	7.7	10.9	15.0	18.1	21.4	21.4	21.9	18.8	15.2	11.1	7.7		
		MCDB	9.8	10.9	15.4	21.4	24.4	28.2	28.1	28.6	23.7	18.1	13.4	9.6		
	5%	WB	5.7	6.5	9.7	13.6	16.9	20.2	20.5	20.7	17.7	14.0	9.8	6.2		
		MCDB	7.5	9.1	13.4	19.3	22.7	26.5	26.9	27.3	21.8	16.5	11.9	7.8		
	10%	WB	4.1	5.1	8.4	12.3	15.7	19.0	19.6	19.7	16.6	12.9	8.3	4.8		
		MCDB	5.6	7.1	11.5	16.9	20.9	24.3	25.4	26.0	19.9	15.1	9.7	6.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	5.7	7.5	9.3	9.1	9.0	9.4	9.4	8.3	6.8	4.5	4.0		
		MCDBR	6.5	8.9	11.9	13.6	13.1	12.8	13.1	13.3	11.8	9.4	7.6	6.9		
	5% WB	MCWBR	4.8	6.1	7.1	7.3	6.5	5.9	5.3	5.7	6.2	5.8	5.3	5.2		
		MCWBR	6.4	8.4	11.1	12.5	12.3	12.4	12.1	12.6	11.0	9.1	7.2	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.321	0.361	0.395	0.404	0.409	0.415	0.408	0.380	0.349	0.322	0.294		
		taud	2.409	2.352	2.310	2.266	2.279	2.303	2.287	2.323	2.374	2.445	2.472	2.453		
	Ebn at Noon		782	834	851	854	859	855	843	833	822	793	743	738		
		Edh at Noon	67	88	108	125	128	126	126	117	101	80	64	57		
(44) All-Sky Solar Radiation	RadAvg		1.10	2.00	3.04	4.22	4.88	5.35	5.26	4.67	3.36	2.20	1.17	0.88		
	RadStd		0.15	0.26	0.36	0.50	0.57	0.47	0.50	0.51	0.44	0.23	0.15	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (102 neighbors)	+0.65	N/A	N/A	+0.83	N/A	N/A	N/A	N/A	N/A	N/A
		+0.51	N/A	N/A	N/A	+0.40	+0.34	N/A	-140	+72	+32

Nomenclature: See separate page